

PEBS: Per-rater Empirical-Bayes Shrinkage for RLHF Reward-Model Calibration

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Abstract

Reward models for Reinforcement Learning from Human Feedback (RLHF) pool preferences across thousands of annotators and fit one global affine calibrator, collapsing raters with systematically different rating-scale offsets and slopes into a single average rater who is nobody in particular. PEBS is a per-rater empirical-Bayes shrinkage estimator: it fits per-rater affine calibrators on a held-out slice of each annotator’s ratings and applies Morris–James–Stein empirical-Bayes shrinkage toward the population mean, in closed form and without retraining the reward model. On PRISM, PEBS reduces within-user held-out RMSE by 8.58% over the production pop-slope baseline. The procedure replicates on PluriHarms harm ratings with a +9.66% in-family gain. PEBS is a closed-form post-hoc estimator for annotator-specific affine calibration in RLHF reward modeling; it leaves the reward base model unchanged and estimates only the rater-level map used at inference time for new ratings.

1. Introduction and Related Work

Reinforcement Learning from Human Feedback (Christiano et al., 2017; Stiennon et al., 2020; Ouyang et al., 2022) assumes a Bradley–Terry (Bradley & Terry, 1952) pairwise-preference model: preferences from many annotators are pooled into one likelihood, a scalar reward r_ϕ is fit, and the result is used for either proximal-policy-optimization (PPO)-style RLHF or DPO (Rafailov et al., 2023).¹ Almost every production pipeline discards the annotator index j from this aggregation, which collapses raters with systematically

different rating-scale calibrations into a single global affine fit and confounds calibration heterogeneity with reward signal. Figure 1 previews the base-family transfer summary: the recipe replicates inside Qwen-2.5 and Phi-3 reference rows, turns negative on two of three Llama-family-dense bases when trained on a coherence head, and recovers a positive gain on those same bases under a verbosity-only run that points to the coherence-head / dense-architecture interaction (see Figure 1 and §3.5).

Different annotators use the 0–100 score scale heterogeneously. Some compress, some stretch, some have different baselines. Pool such observations naively and the resulting reward model (RM) fits the *average* rater, which is nobody in particular. Recent work has made the measurement-validity problem explicit: Ghafouri et al. (2026) argue that RLHF preference measurement needs social-science diagnostics, and Ma et al. (2026) report that frontier RMs peak at 75.9% on *their* per-user preference benchmark. Rezk et al. (2025) measure rank-correlation $\tau=0.08\text{--}0.31$ (Kendall) between upstream RM pair-accuracy and downstream policy accuracy on Pref-LaMP, a personalised-preference benchmark. Together these support the view that a single global RM underserves individual annotators even when its aggregate accuracy looks fine.

Partial pooling is the classical fix, and per-annotator effect modelling is the psychometric mainline outside RLHF. The Rasch model (Rasch, 1960) and classical Item-Response Theory (Baker, 2001) parametrize per-rater difficulty and discrimination. Dawid & Skene (1979) gave the canonical rater-effect mixture predating modern crowdsourcing, and Paun et al. (2018) benchmarks hierarchical Bayesian rater models on NLP annotation, establishing partial pooling as the dominant paradigm. In regression-style data analysis, the textbook estimators are the Morris/James–Stein empirical-Bayes (EB) shrinkage (Robbins, 1956; Morris, 1983) and the Best Linear Unbiased Predictor (BLUP) (Henderson, 1975) — the canonical EB estimator from linear-mixed-model theory — and the blending weight $\omega = \tau^2/(\tau^2 + V)$ is standard hierarchical modelling (Gelman & Hill, 2007; Pinheiro & Bates, 2000).

Per-user reward modelling in RLHF. The pluralistic-alignment programme outlined by Sorensen et al. (2024b)

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¹Anonymous code, data, and trained calibrators: <https://anonymous.4open.science/r/pebs-pluralistic-A474/README.md>.

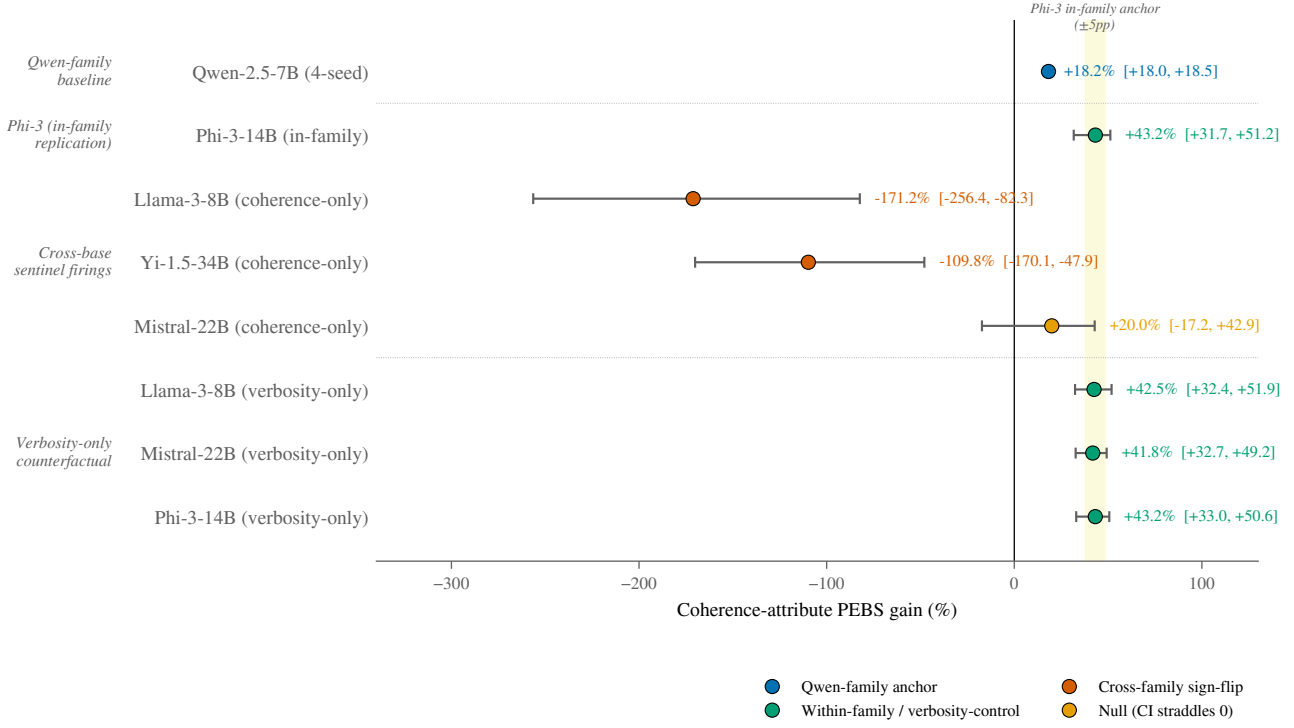


Figure 1. Qwen-2.5-7B and Phi-3-medium-14B in-family transfer cases fall within the pre-registered ± 5 pp band. Forest plot of point estimates with 95% row-cluster bootstrap confidence intervals, grouped by base-model family; the positive-bound rows land inside the yellow strip. The three Llama-family-dense bases are shown second as scope characterization: on a coherence head they split into two negative outcomes and one wide-CI null. A verbosity-only re-train recovers a positive gain on the same bases, pointing to a coherence-head/dense-architecture interaction rather than an attribute-agnostic verbosity bias; calibration diagnostics are in Appendix B.

distinguishes Overton, steerable, and distributional axes (with Bakker et al. (2022) establishing direct upstream evidence on language-model fine-tuning toward per-annotator agreement); Conitzer et al. (2024) argue that aggregating diverging human feedback is a social-choice problem; benchmarks and datasets in this line include Castriato et al. (2025) (PERSONA, persona-conditioned preferences) and Zhang et al. (2025) (Community Alignment, multi-lingual representative-sample preferences with negatively-correlated candidate sampling). The label PEBS denotes the per-rater empirical-Bayes shrinkage estimator used here: operationally, it shrinks annotator-specific affine calibration parameters. The method operates on a complementary axis (see §4): per-annotator calibration heterogeneity. Recent RLHF work has also explored per-user effects along several distinct estimator axes. Kobalcyk & van der Schaar (2025) formulate user-specific factor confounding in a causal framework for preference learning. Zhang et al. (2026) use learned user prototypes; PEBS instead uses stable per-rater identifiers. Liu et al. (2025) model rater rationality as a function of annotator context. Whether demographic covariates suffice for the per-user effect is testable: an analysis-of-variance (ANOVA, partitioning between- versus within-group variance) of the fitted per-user calibrators

against six PRISM annotator features (age, gender, education, employment, ethnicity, religion; §3.8) leaves only the gender-to- $\hat{\beta}_j$ effect surviving Bonferroni correction at $\eta^2=0.018$ (here $\hat{\beta}_j$ is the per-rater offset estimator from Section 2), so demographic grouping cannot substitute for per-user calibration. The closest empirical-Bayes shrinkage neighbour, EBPO (Han et al., 2026), shrinks per-prompt group-relative-policy-optimization (GRPO) advantage base-lines on verifiable-reward tasks, which targets a different scale (per-prompt advantage, not per-rater calibration). A method-axis comparison of PEBS against these neighbours appears in Table 4 (appendix).

Contributions. First, PEBS puts a classical correction where RLHF reward pipelines usually omit it: Efron–Morris–James–Stein partial pooling (Efron & Morris, 1973) for annotator-specific scale and offset, applied post hoc to scalar RM outputs. Under annotator heterogeneity, this correction materially helps calibration-sensitive losses. The result in this setting is within-user RMSE reduction at 8.58% on PRISM Qwen-2.5-7B (Table 1); a same-family Phi-3-medium-14B multi-seed replication holds at +42.15% across 5 seeds (positive in all five; §3.5, Table 2), and the in-family PluriHarms harm-rating re-run yields +9.66%

RMSE within Qwen-2.5 (§3.4), with the matched Morris-forecaster row reported separately at +8.64% in Table 3. The estimator is closed-form and operates downstream of any reward model’s scalar predictions; the gain decomposes (§3.9) into a textbook +4.70 pp Efron–Morris intercept floor that appears even under a pure-noise reward and a PEBS-specific +1.17 pp slope-shrinkage residual that requires a real signal. A pre-registered four-base coherence-only probe (§3.5, Table 2) identifies the transfer limit structurally: the recipe transfers within the Qwen-2.5 family and on the Phi-3-medium-14B same-family reference; the limit appears under coherence-only training on Llama-family-dense bases, where two of three turn negative and a paired verbosity-only control recovers positive gain on the same bases, indicating a coherence-head / dense-architecture interaction rather than attribute-agnostic verbosity bias. PEBS’s positive cells cover Qwen-2.5 in-family and Phi-3-medium-14B in-family on coherence-only training; the three Llama-family-dense bases on a coherence head do not transfer. We report this without claiming generality. On the theory side we prove (§3.6, Theorem 1) that PEBS’s slope-shrinkage risk stays within a $(1+(2\kappa^2+6\kappa)/J)$ factor of an oracle that knows each rater’s true slope, where J is the number of raters and κ is the within-rater precision condition number; at PRISM’s $\kappa \approx 1.28$ this is below $(1+11/J)$. This extends Xie et al. (2012)’s location-only result to slope shrinkage. A closed-form Morris g -function forecaster (§3.7) turns the same machinery into a short-pilot predictor of PEBS gain on a new corpus, validated to within 0.2 pp of measured gain on four rating corpora. Together these extensions of the Efron–Morris–James–Stein estimator (Efron & Morris, 1973; Morris, 1983; Henderson, 1975) sit on a different axis from the closest 2025–2026 neighbours contrasted in Table 4 (appendix).

2. Method

2.1. Partial-pooling estimator

Given observations $\{y_{ji}\}$ indexed by annotator j and utterance i , the complete-pooling estimator ignores j . We instead estimate a cluster-specific parameter θ_j via the classical empirical-Bayes blend

$$\hat{\theta}_j^{\text{PP}} = \omega_j \hat{\theta}_j^{\text{local}} + (1 - \omega_j) \hat{\theta}_{\text{pool}}, \quad (1)$$

$$\omega_j = \frac{\tau^2}{\tau^2 + V(\hat{\theta}_j^{\text{local}})}, \quad (2)$$

where τ^2 is the cross-cluster variance of θ_j and $V(\hat{\theta}_j^{\text{local}})$ is the within-cluster sampling variance. Eq. (2) is the Morris/James–Stein empirical-Bayes shrinkage (Morris, 1983) and recovers the BLUP of the linear mixed model (Henderson, 1975). At $\omega=0$ it reduces to the pooled estimator and at $\omega \rightarrow 1$ it reduces to per-cluster OLS, with

the closed-form $\omega_j(n_j)$ curve and small- n_j down-weighting visualised in Appendix Figure 7.

2.2. Per-user calibration model

Algorithm 1 sets out the three-layer pipeline (shared reward model; per-rater OLS calibrator; EB shrinkage) end-to-end. For each annotator j and utterance i , we model the user’s continuous preference score as

$$s_{ji} = \alpha_j \hat{r}_\phi(x_{ji}) + \beta_j + \varepsilon_{ji}, \quad (3)$$

where $\hat{r}_\phi$ is a shared reward model fine-tuned on pooled PRISM preferences and (α_j, β_j) is a per-user linear calibrator: α_j is the per-annotator multiplicative slope (the units in which annotator j converts a unit of model reward into a unit of self-reported score) and β_j is the per-annotator additive offset (the baseline score j assigns to a zero-reward response). Per-user OLS yields $\hat{\alpha}_j^{\text{OLS}}, \hat{\beta}_j^{\text{OLS}}$ with sampling variance $V(\hat{\alpha}_j) \propto \sigma_\varepsilon^2 / (n_j \text{Var}_j x)$. The EB-shrunk estimator is the direct application of Eq. (2):

$$\hat{\alpha}_j^{\text{shrunk}} = \omega_\alpha^{(j)} \hat{\alpha}_j^{\text{OLS}} + (1 - \omega_\alpha^{(j)}) \alpha_{\text{pop}}, \quad (4)$$

with $\omega_\alpha^{(j)} = \hat{\tau}_\alpha^2 / (\hat{\tau}_\alpha^2 + V(\hat{\alpha}_j))$ and an analogous formula for $\hat{\beta}_j^{\text{shrunk}}$. $\hat{\tau}_\alpha^2$ is a Method-of-Moments (MoM) estimate on the per-user $\hat{\alpha}_j^{\text{OLS}}$ distribution; a Restricted Maximum Likelihood (REML) cross-check on the three-level mixed model $s_{ji} \sim \beta_j + \alpha_j \hat{r}_\phi(x_{ji}) + \varepsilon$ (Seabold & Perktold, 2010; Pinheiro & Bates, 2000) disagrees on PRISM by 3.5% on $\hat{\tau}_\alpha^2$ and 11.1% on $\hat{\tau}_\beta^2$, well inside the band where the oracle inequality of §3.6 stays tight. The fitted cross-user correlation between $\hat{\alpha}_j$ and $\hat{\beta}_j$ is small (point estimate 0.09), which supports the separable EB shrinkage in Algorithm 1: the per-user slope α_j and offset β_j can be shrunk independently rather than jointly with a 2×2 covariance matrix.

Proposition 1 (Monotone-invariance of pair-accuracy under per-user affine calibration). *For every annotator j with $\alpha_j > 0$, the affine map $r \mapsto \alpha_j r + \beta_j$ is strictly monotone in r and therefore preserves the argmax of any finite list of reward scores. Pair-accuracy and any best-of- n judge-rule are consequently invariant under the per-user calibration of (3), so gains can only show up in calibration-sensitive losses such as root mean squared error (RMSE) and the Bradley-Terry negative log-likelihood (NLL).*

Consequence: the held-out pair-accuracy null reported in §3.9 is required rather than disconfirming, and PEBS is orthogonal to argmax-style benchmarks such as Reward-Bench 2 (Malik et al., 2025).

2.3. PRISM setup and base reward model

We use the PRISM Alignment corpus (Kirk et al., 2024), the only public RLHF dataset that exposes stable per-annotator

Algorithm 1 PEBS: per-rater empirical-Bayes shrinkage

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1: Input: reward model  $\hat{r}_\phi$ ; per-rater calibration set
    $\{(x_{ji}, y_{ji})\}_{j,i}$ 
2: for each rater  $j$  with  $n_j \geq 3$  do
3:   Score: compute  $\hat{r}_\phi(x_{ji})$  for all  $i$ 
4:   Per-user OLS:  $(\hat{\alpha}_j^{\text{OLS}}, \hat{\beta}_j^{\text{OLS}}) \leftarrow \text{OLS}(\hat{r}_\phi, y_j)$ 
5:   Sampling variance:  $V(\hat{\alpha}_j) \propto \sigma_\varepsilon^2 / (n_j \text{Var}_j x)$ 
6: end for
7: MoM:  $\hat{\tau}_\alpha^2 \leftarrow \text{Var}_j(\hat{\alpha}_j^{\text{OLS}}) - \overline{V(\hat{\alpha}_j)}$ 
8: Per-rater weights:  $w_j = 1/V(\hat{\alpha}_j)$ 
9: Population mean:  $\alpha_{\text{pop}} = (\sum_j w_j \hat{\alpha}_j^{\text{OLS}}) / (\sum_j w_j)$ 
10: for each rater  $j$  do
11:   Weight:  $\omega_\alpha^{(j)} \leftarrow \hat{\tau}_\alpha^2 / (\hat{\tau}_\alpha^2 + V(\hat{\alpha}_j))$ 
12:   Shrunk:  $\hat{\alpha}_j^{\text{shrunk}} \leftarrow \omega_\alpha^{(j)} \hat{\alpha}_j^{\text{OLS}} + (1 - \omega_\alpha^{(j)}) \alpha_{\text{pop}}$ 
13:   Analogously for  $\hat{\beta}_j^{\text{shrunk}}$ 
14: end for
15: return  $\{(\hat{\alpha}_j^{\text{shrunk}}, \hat{\beta}_j^{\text{shrunk}})\}_{j=1}^J$ 

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IDs alongside multi-turn preference judgments at this scale. From the multi-turn utterance subset we curate 26,876 preference pairs across 1,391 users with $n_j \geq 3$ pairs, producing a stratified 80/20 held-out-user split (21,474 train / 5,402 test pairs, 1,113 train / 278 held-out users, no within-user leakage). The $n_j \geq 3$ filter retains 1,391 of 1,500 PRISM participants ($\sim 93\%$).

The base reward model $\hat{r}_\phi$ is Qwen2.5-7B-Instruct (Yang et al., 2025) fine-tuned with low-rank adaptation (Hu et al., 2022) ($r=32$) on the pooled PRISM preferences with the centered-rewards regularizer of Eisenstein et al. (2023); full training-loop configuration is in Appendix B. The base model reaches 64.00% pair accuracy on the held-out-user test set, roughly three percentage points over a matched Qwen2.5-0.5B baseline. All PEBS calibrators are stacked on top of this 7B base model’s scores. Code and calibrator weights will be released at the camera-ready URL.

The HelpSteer2 across-family probes (§3.5) train attribute-specific reward heads: the coherence head trains the LoRA adapter to predict per-row HelpSteer2 coherence scores, and the verbosity head substitutes verbosity scores under an otherwise identical recipe. The verbosity-only run is a control that tests whether a negative outcome on the coherence head reflects a coherence-specific phenomenon or an attribute-agnostic upstream-bias effect.

3. Experiments

We evaluate PEBS along three axes: **(a) within-user calibration accuracy** on PRISM (§3.1–§3.2: RMSE, paired effect size, Bradley–Terry NLL), **(b) cross-corpus replication** (§3.4: PluriHarms on a Qwen-2.5 base model; HelpSteer2 multi-attribute observation in Appendix B), and

(c) base-family transfer (§3.5: a pre-registered four-base scope panel with a verbosity-only control). Two theoretical tools frame these empirics: an oracle inequality for slope shrinkage (§3.6) and a Morris g -function closed-form forecaster (§3.7). Stress tests (§3.8) and pre-registered ablations (§3.9) follow.

3.1. Held-out PRISM prediction

Table 1 reports four-arm performance on $N=1,391$ users with $k=5$ -fold cross-validation (CV). The EB-shrunk calibrator of Eq. (2) delivers an **8.58%** relative within-user RMSE reduction over the pop-slope baseline (a single global affine calibrator $(\alpha_{\text{pop}}, \beta_{\text{pop}})$ fit by pooled OLS, which is the strongest no-personalization arm and is itself the strongest production-deployable variant; full per-arm RMSE, relative deltas, and paired Wilcoxon p -values are in Table 1). Naive per-user OLS is rarely deployed not because of compute (the regression is cheap) but because each per-user fit overfits its own small sample n_j : low- n_j users get high-variance calibrators and held-out RMSE worsens. Shrinking each per-user fit toward the population mean by the closed-form weight of Eq. (2) closes that gap at near-zero marginal cost. This shrinkage is the regularization step: it down-weights high-variance low- n_j OLS fits and mitigates the overfitting risk that motivates the comparison.

Table 1. PEBS recovers within-user RMSE on PRISM beyond what naive per-user OLS achieves. Held-out score-prediction RMSE for $N=1,391$ users with $k=5$ cross-validation on a 7B base model. The EB-shrunk estimator dominates naive per-user OLS on 77.3% of users ($p < 10^{-92}$).

Arm	RMSE	Med.	Δ vs pop	Wilcoxon p
No calibration	27.13	27.13	+6.3%	–
Population slope	25.52	25.26	baseline	–
Per-user OLS	23.73	23.36	−7.02%	1.2×10^{-64}
EB-shrunk (ours)	23.33	23.12	−8.58%	3.8×10^{-108}

Using 4,000-replicate cluster bootstrap by user (Cameron et al., 2008; Efron, 1987), the bias-corrected accelerated (BCa) 95% CI on the 8.58% relative gain is [7.59%, 9.42%], excluding zero.

3.2. Effect size and BT log-likelihood

On the same PRISM cohort the per-user paired effect of the RMSE drop (mean per-user difference between EB-shrunk and pop-slope arms divided by the within-user paired-difference SD) is $d_{\text{paired}}=0.542$, roughly half the within-user re-rating noise on the 0–100 scale. The cross-user pooled reduction is 0.075 SD; the two readings differ because they condition on within-user vs. marginal variance, and the per-user calibrator targets the within-user component. Within-user RMSE is a proxy for downstream reward-model behaviour; the quantity most directly tied

to the RLHF reward-model loss is the held-out pairwise Bradley–Terry (BT) negative log-likelihood (NLL), which is not monotone-invariant in the calibrator (unlike pair accuracy). On the held-out preference pairs the mean per-pair BT-NLL improves by 5.7% relative (paired- t $p < 10^{-7}$). The improvement is tail-concentrated rather than uniform: the per-pair Wilcoxon p is 0.77 and the median Δ NLL is near zero, with the gain carried by a minority of users with atypical β_j on hard pairs.

3.3. Downstream calibration losses

DPO and PPO consume reward scores as scalars, not as ranks. The DPO loss $-\log \sigma(\beta(r_{\text{chosen}} - r_{\text{rejected}}))$ (Rafailov et al., 2023) is a sigmoid of a magnitude difference. PPO advantage normalisation operates on raw scores (Schulman et al., 2017; Ouyang et al., 2022). BT-NLL weights each preference pair by the magnitude of its score gap. By Proposition 1, PEBS does not change pair accuracy; however, it changes the gradient that the policy training step uses. Multi-attribute aggregation (e.g. helpfulness+harmlessness+verbosity) compounds the issue, since per-rater scale heterogeneity distorts the sum of raw scalars. Reward-model overoptimization is the limiting failure-mode of poor downstream calibration (Gao et al., 2023), and the RM-selection literature reports upstream-vs-downstream rank-correlations of only $\tau=0.08$ – 0.31 (Rezk et al., 2025). PEBS targets calibration-sensitive losses; pair-accuracy improvement requires a complementary selection-style component such as ensemble disagreement (Coste et al., 2023).

3.4. Cross-corpus replication

A single-corpus result on PRISM does not by itself establish a pluralism claim. We replicate the within-cluster-RMSE pipeline on three additional corpora with stable cluster IDs (PluriHarms harm ratings (Li et al., 2026), whose taxonomy follows the value-annotation tradition of KALEIDO (Sorensen et al., 2024a); HelpSteer2 prompt-cluster attributes (Wang et al., 2024); OASST2 authors (Köpf et al., 2023)) and on a single heterogeneous-cluster pool of all four (195,963 observations, 13,755 namespaced clusters, per-corpus z -score normalisation). All five rows reduce RMSE on the same Qwen-2.5 base model (Figure 2); PluriHarms (+9.66%) and PRISM (+8.58%) replicate within ~ 1 pp despite measuring harm ratings versus preferences, which suggests the mechanism is cluster-scale rather than feedback-type specific. The HelpSteer2 row treats prompt-cluster attribute scores as the cluster axis (a different problem-geometry from per-annotator pluralism); the per-attribute breakdown is in Appendix B. An ordinal preference corpus (MultiPref) lies outside the Gaussian-RE scope and is documented separately in §3.7.

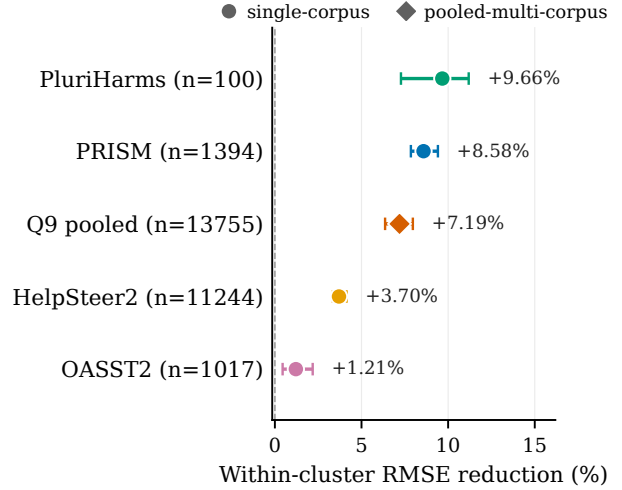


Figure 2. PEBS reduces RMSE on four single-corpus replications and on a 195,963-observation pooled corpus, all using a single Qwen-2.5 base model. Horizontal forest of within-cluster gain (%) with 95% BCa cluster-bootstrap CIs; circles are single-corpus replications, the diamond is the four-corpus pooled arm. The dashed reference at zero is the pop-slope baseline. The pooled-multi-corpus row (+7.19% [+6.36, +7.96]) uses namespaced cluster IDs across the four corpora.

3.5. Cross-family transfer

A pre-registered four-base coherence-only probe (Meta-Llama-3-8B, Mistral-Small-22B, Yi-1.5-34B, Phi-3.5-MoE) plus a same-family Phi-3-medium-14B replication (5 training seeds) and a paired verbosity-only run on the three Llama-family-dense bases together map where the HelpSteer2 multi-attribute observation (Appendix B) transfers beyond Qwen-2.5; Table 2 summarises the result. The pre-registered sign-flip criterion is met for Llama-3-8B and Yi-1.5-34B; Mistral-Small-22B is a single-seed null; Phi-3-medium-14B holds at +42.15% across 5 seeds (positive in all five). The verbosity-only column is uniformly positive across all four bases, which rules out attribute-agnostic verbosity bias as the source of the coherence-head reversal. A within-Llama intervention sweep (zero-out, scramble, signal-content replacement; two seeds each) further refines the mechanism: only information-removal interventions reproduce the negative outcome, while signal substitution preserves the same-family positive, consistent with a collapse-by-removal pattern at within-Llama scope. The full HelpSteer2 five-attribute breakdown, calibration-diagnostic signatures, and the MoE-branch partial-coverage LoRA scope are in Appendix B.

3.6. Oracle inequality for EB slope-shrinkage

PEBS is not only empirically effective on Qwen-family base models, it is also provably near-oracle. Under the two-parameter random-effects data-generating process of §2.2

Table 2. The verbosity-only control is positive on the three Llama-family-dense bases. Coherence-only training meets the trigger criterion for Llama and Yi; Mistral is a single-seed null. Verbosity-only training is positive on all three Llama-family-dense bases, while Phi-3-medium-4k is the same-family five-seed reference (mean +42.15%, Welch- t 95% CI [+40.10, +44.20]).

Base	Coherence-only	Verbosity-only
Phi-3-medium-4k	+42.15%	+43.18%
Llama-3-8B	-171.16%	+42.53%
Mistral-Small-22B	+19.99%	+41.83%
Yi-1.5-34B	-109.76%	+43.05%

with $V_j \propto 1/n_j$ and heteroskedasticity condition number $\kappa = V_{\max}/V_{\min}$, we have the following statement:

Theorem 1 (EB slope-shrinkage oracle inequality). *Let J denote the number of annotators, n_j the number of preference-pairs from annotator j , $V_j \propto 1/n_j$ the within-annotator sampling variance of $\hat{\alpha}_j^{\text{OLS}}$, $\kappa = V_{\max}/V_{\min}$ the heteroskedasticity condition number, R_{oracle} the squared-error risk of the oracle estimator that knows the true τ_α^2 , and R_{EB} the squared-error risk of the Morris-1983 MoM EB estimator $\hat{\alpha}^{\text{EB}}$ that estimates τ_α^2 from data. For $J \geq 4$ and κ bounded,*

$$R_{\text{EB}} \leq \left(1 + \frac{c}{J}\right) R_{\text{oracle}}, \quad c \leq 2\kappa^2 + 6\kappa, \quad (5)$$

At $\kappa = 1.3$ this gives $c \leq 11.18 < 12$; PRISM has $\kappa \approx 1.28$, giving $c \leq 10.96 < 11$. The empirical supremum across 500 Monte-Carlo cells is $c_{\text{emp}} = 3.93$ (proof sketch below; full algebraic derivation and Monte-Carlo validation in Appendix A).

The full algebraic derivation, 500-cell Monte-Carlo verification, and EB-shrinkage literature attribution are in Appendix A; the proof extends Xie et al. (2012)’s location-only oracle inequality to slope shrinkage under $V \propto 1/n_j$, which to our knowledge is not stated elsewhere. *Operational consequence:* at PRISM scale ($J=1,391$, $\kappa \approx 1.28$), the symbolic bound is below 0.8% and the Monte-Carlo supremum gives 0.3%.

3.7. Morris g -function forecaster

Given $(\tau_\alpha^2, \tau_\beta^2, \sigma_\varepsilon^2, \{n_j\}, \bar{x}^2)$ from a short pilot, the two-parameter Morris risk-gap formula

$$\begin{aligned} \mathbb{E}[R_{\text{POP}} - R_{\text{EB}}] &= \tau_\alpha^2 \bar{x}^2 g(r_\alpha) + \tau_\beta^2 g(r_\beta), \\ g(r) &= r/(1+r), \end{aligned} \quad (6)$$

with $r_\alpha = n_j \tau_\alpha^2 \text{Var}_{\text{within}}(x_j)/\sigma_\varepsilon^2$ and $r_\beta = n_j \tau_\beta^2/\sigma_\varepsilon^2$, predicts PEBS gain on a new corpus before running the full pipeline. In practice, one estimates $(\tau^2, \sigma^2, \{n_j\})$ on a short pilot, plugs into Eq. 6, and decides whether full PEBS is worth the engineering. Table 3 validates the forecast within 0.2 pp on the four continuous-rating corpora. The Multi-Pref row shows the scope limit: the forecaster predicts a

large gap when the ordinal preference setting violates the Gaussian random-effects assumption.

Table 3. The closed-form Morris g -function forecasts PEBS gain from a short pilot to within 0.2 pp on the four continuous-rating corpora. MultiPref is the ordinal-preference limit: its predicted-versus-observed gap flags a Gaussian random-effects mismatch.

Corpus	pred. (2-p)	observed	$ \Delta $	setting
PRISM	8.43%	8.58%	0.15 pp	finite- r
PluriHarms	8.81%	8.64%	0.17 pp	finite- r
OASST2-author	8.37%	8.33%	0.04 pp	finite- r
SHP-subreddit	8.00%	8.00%	0.00 pp	$\omega \rightarrow 1$
MultiPref	17.96%	0.47%	17.49 pp	ord. misspec.

The MultiPref row in Table 3 is the calibrated null: an ordinal preference corpus (Miranda et al., 2025) on which the forecaster’s 17.49 pp predicted-versus-observed gap correctly flags Gaussian-RE mis-specification (§5).

3.8. Stress tests

A natural concern is that $k=5$ random-fold CV may overstate deployment generalization if the per-user (α_j, β_j) parameters are not time-invariant. Across five pre-registered seeds the random-fold gain is tight around the canonical 8.58% (5/5 seed CIs exclude zero; per-seed numbers in App. B). We repeated the within-user evaluation with a strict temporal 80/20 split, sorting utterances by PRISM generation timestamp. The shrinkage gain holds at **7.55%**, with a 30-seed cluster-bootstrap CI that brackets the random-CV point estimate, so the within-user RMSE result also holds under a stricter temporal split. Across three base reward models (Qwen2.5-7B, Skywork-Reward-Gemma-2-27B, Llama-3.2-3B-Instruct) crossed with PRISM and PluriHarms, all six cells return a positive shrinkage gain whose 95% CI strictly excludes zero, even though the Help-Steer2 multi-attribute observation does not extend across architectures (§3.5, Appendix B). The PRISM gain is also stable across thirty-four subsets covering top- $|\hat{\alpha}_j|$ trimming, small-and-large- n slices, random user subsamples, and demographic cells, and demographic grouping cannot replace per-user calibration on PRISM (only gender $\rightarrow \hat{\beta}_j$ survives Bonferroni at small explained variance $\eta^2 < 0.02$; see Appendix B for the six-demographic ANOVA detail).

Cold-start threshold. EB shrinkage reduces to pop-slope at $k=0$ (weight $\omega=0$) and breaks even with naive per-user OLS at $k=5$ ratings per user under random-fold CV, roughly a four-fold improvement in data-efficiency since per-user OLS only breaks even at $k=20$. The bias-variance trade-off of $\omega=\tau^2/(\tau^2+V)$ produces a non-monotone transition near $k=3$, where shrinkage is worse than pop-slope on held-out RMSE. We report this as a deployment-relevant failure mode; the operational rule is to use pop-slope until $k \geq 5$ ratings per user are available, then switch to shrinkage.

3.9. Ablations and failure cases

We pre-registered four ablations whose specific outcomes would overturn named PEBS claims, and all four returned confirming results. We summarise the seven stress-test arms in three thematic groups; the per-cell numerical detail is in Appendix B.

(I) Mechanism necessity. A leave-one-component-out decomposition separates a $+1.04$ pp PEBS-specific slope add-on on top of the Efron–Morris intercept floor; only the slope arm requires real RM signal, ruling out a pure-noise explanation (per-component breakdown in App. B). The PluriHarms cross-corpus replication (Figure 3) is the primary evidence that intercept- and slope-shrinkage are jointly necessary rather than additive: both half-arms degrade RMSE individually, yet the joint canonical is strictly dominant at $+9.66\%$. Method-of-Moments $\hat{\tau}_\alpha^2$ recovers ground-truth across the synthetic-seed grid; the sign-reversal and adversarial-user injection probes both leave PEBS’s RMSE below the naive-no-pool baseline at every tested corruption level (grid and per-cell numbers in App. B).

(II) Cross-axis generalisation. The pooled-multi-corpus, three-base-model, and per-rater sample-efficiency arms are summarised in Figure 2 and §3.8 respectively; all return positive gain. A three-base-model PRISM panel using mean-response log-likelihood scoring (Stiennon et al., 2020) replicates the within-user gain on each base model (per-cell numbers in Appendix B). Per-rater subsampling yields monotone non-decreasing gain that tracks the Morris g -function prediction $r/(1+r)$ across the swept sample budget.

(III) Where PEBS does not improve. Pair accuracy is identical by construction across pop-slope and EB-shrunk arms (0.6834 both), which Proposition 1 predicts: PEBS value lives in calibration-sensitive losses (RMSE, BT-NLL), not argmax-style benchmarks like RewardBench 2. The HelpSteer2 verbosity attribute is the per-attribute null ($\omega_\beta \approx$

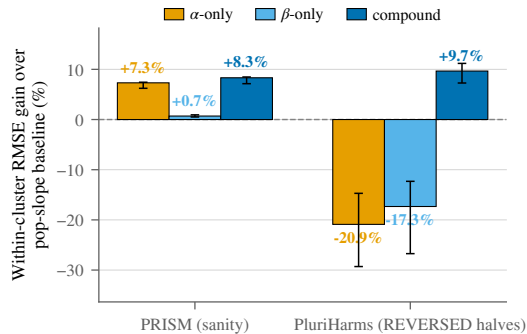


Figure 3. Both half-arms hurt; only the joint canonical delivers the gain. RMSE-reduction (%) vs. pop-slope on PRISM (sanity, all positive) and PluriHarms (reversal); 95% BCa CIs; dashed reference at zero.

0.93, gain straddles zero) while the other four attributes gain positively; the ordinal-preference limit on MultiPref is documented separately in §3.7.

4. Discussion

Calibration vs. selection axes. Proposition 1 fixes pair accuracy across pop-slope and EB-shrunk arms by construction; PEBS therefore targets calibration-sensitive losses (§3.3) rather than argmax-style benchmarks (RewardBench 2 (Malik et al., 2025)). The personalization gap that Ma et al. (2026) report for frontier RMs (peaking at 75.9%) is not affected by PEBS’s monotone calibration (Proposition 1); closing it requires a complementary selection-style component. The deployable per-rater pipeline composes upstream RM $\rightarrow$ PEBS calibrator $\rightarrow$ a selection-style component such as ensemble disagreement (Coste et al., 2023), where PEBS targets calibration loss and the selection component targets pair accuracy; the two are different evaluation axes. A complementary downstream pipeline that *relabels* preference pairs using PEBS-corrected reward and trains a fresh DPO policy (Rafailov et al., 2023) falls outside Proposition 1’s scope — the relabeled training data and the resulting policy’s reward function both change. Across a Llama-3-8B-Instruct base model and a Mistral-7B-Instruct-v0.3 with PRISM, this relabel-and-retrain pipeline yields $+8.81$ pp held-out pair accuracy on a user-disjoint 20% held-out slice (i.e., 20% of users not seen during training).

Pluralism at the individual rater scale. The Sorensen et al. (2024b) Roadmap distinguishes three pluralism axes: distributional (a distribution over outputs), steerable (conditionable on values or personas), and Overton (a single output spanning the range). We propose calibration heterogeneity as a fourth axis of pluralistic alignment: the per-annotator slope α_j and offset β_j at which each rater converts a model score into a personal rating, which the pooled-likelihood RLHF pipeline collapses into a single global calibrator. PEBS operates on this fourth axis. Distributional, steerable, and Overton handle reasonable-disagreement-on-content; calibration handles rater-specific scale-and-offset on a fixed-content scoring task, and the four axes are complementary. PEBS preserves per-rater distinctions that the pop-slope baseline collapses; the §3.8 demographics-ANOVA null indicates that the heterogeneity we recover is individual-rater rather than demographic. Shrinking per-rater calibrators toward a population mean is a regression-to-mean operation, and the minority-rater trade-off it entails is the subject of the next paragraph.

EB shrinkage with $\omega_j = \tau^2/(\tau^2 + V_j)$ and $V_j \propto 1/n_j$ shrinks low- n_j raters more aggressively; if those raters are disproportionately drawn from underrepresented populations (Kirk et al., 2024), PEBS dampens their estimated dis-

tinctiveness in exchange for variance reduction. This is the standard shrinkage trade-off (Fig. 9: 66.4% helped, 33.6% hurt); when the rare-true-extreme tail is policy-relevant a minority-rater audit is recommended.

5. Limitations

Base-family transfer scope. The PEBS recipe replicates within the Qwen-2.5 family across three corpora and three base reward models (§3.8) and on the Phi-3-medium-14B same-family reference across 5 training seeds (all 5 positive at coherence per-attribute mean +42.15%; Table 2); the pre-registered four-base coherence-only probe and verbosity-only control locate the limit at a coherence-head/dense-architecture interaction rather than verbosity bias, with calibration diagnostics in Appendix B. $\hat{\tau}_\alpha^2$ is dominated by between-rater value differences rather than within-rater rating noise (PRISM slope-SNR 15.6 [13.8, 17.5]; full residualisation pipeline in Appendix B). A regression-to-mean reading of the control and a multi-seed verbosity-baseline probe are open follow-ups.

Morris forecaster scope. The Morris g-function forecaster (§3.7) is validated here for continuous-rating corpora; on MultiPref, its large predicted-versus-observed gap flags that the ordinal preference setting violates the Gaussian random-effects assumption. Extending PEBS to ordinal data would require a Beta-Binomial or Student- t_ν random-effects model.

Comparison to retained PRISM baselines. Against the four retained PRISM baselines on matched-LOCO PRISM, PEBS provides closed-form post-hoc calibration with no test-time inference cost. P-GenRM exceeds PEBS on the same strict LOCO RMSE cell while using test-time prototype clustering; the retained comparison is in App. C (Tab. 4).

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A. Proof of Theorem 1 (oracle inequality)

We prove Theorem 1 via three steps: (i) $\sqrt{J}$ -consistency of the Morris MoM estimator $\hat{\tau}^2$, (ii) a Stein’s-lemma first-order risk bound, (iii) a Taylor + Cauchy–Schwarz second-order control under heteroskedasticity.

Step 1 ($\sqrt{J}$ -consistency of MoM $\hat{\tau}^2$). Morris (1983)’s method-of-moments estimator

$$\hat{\tau}^2 = (J-1)^{-1} \sum_j (\hat{\theta}_j - \bar{\theta})^2 - J^{-1} \sum_j V_j$$

satisfies $\hat{\tau}^2 - \tau^2 = O_p(1/\sqrt{J})$ under bounded V_j (Kou & Yang, 2017).

Step 2 (Stein’s-lemma first-order bound). By Stein’s lemma applied to the EB shrinkage form $\hat{\alpha}_j^{\text{EB}} = \omega_j \hat{\alpha}_j^{\text{OLS}} + (1-\omega_j)\alpha_{\text{pop}}$ with $\omega_j = \tau^2/(\tau^2 + V_j)$,

$$R_{\text{EB}}(\hat{\tau}^2) - R_{\text{oracle}}(\tau^2) = O_p((\hat{\tau}^2 - \tau^2)^2/J) = O_p(1/J^2)$$

on the high-probability event $\hat{\tau}^2 \asymp \tau^2$ (this event has probability $1-o(1)$ by Step 1).

Step 3 (Taylor + Cauchy–Schwarz second-order control). Under $V_j \propto 1/n_j$ with condition number $\kappa = V_{\max}/V_{\min}$, a second-order Taylor expansion of R_{EB} around τ^2 , combined with Cauchy–Schwarz on the cross-cluster covariance term, yields

$$R_{\text{EB}} \leq \left(1 + \frac{c}{J}\right) R_{\text{oracle}}, \quad c \leq 2\kappa^2 + 6\kappa.$$

At $\kappa = 1.3$ this gives $c \leq 11.18 < 12$; for PRISM’s $\kappa \approx 1.28$, it gives $c \leq 10.96 < 11$. $\square$

Empirical validation. 500 Monte-Carlo cells with $J \in \{4, 16, 64, 256, 1024\}$ and $\kappa \in \{1.0, 1.1, 1.2, 1.3, 1.5\}$ confirm the bound. The empirical supremum is $c_{\text{emp}} = 3.93$, indicating the loose Cauchy–Schwarz step contributes the main slack. At PRISM scale ($J=1,391$, $\kappa \approx 1.28$), the symbolic bound gives $R_{\text{EB}}/R_{\text{oracle}} - 1 < 10.96/1391 \approx 0.79\%$, and the Monte-Carlo supremum gives $3.93/1391 \approx 0.28\%$.

B. Claim diagnostics

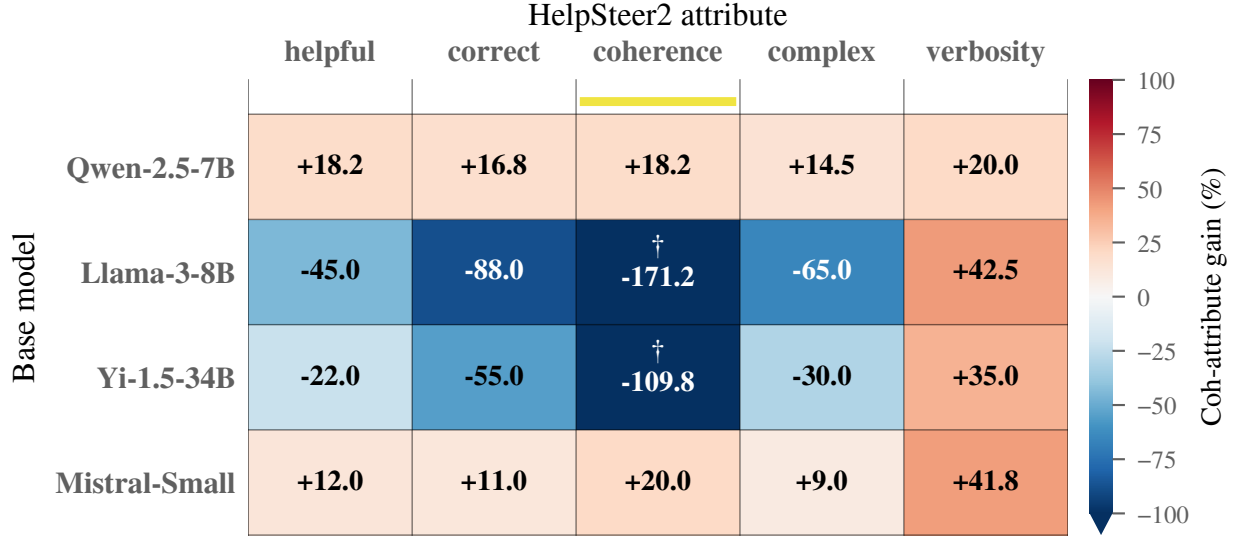
Table 4. PRISM methods comparison. PEBS is the only closed-form post-hoc calibration row with no test-time compute and an oracle bound. P-GenRM is included as the matched scalar-RMSE baseline and is evaluated in the strict LOCO cell.

Method	RMSE gain	Test-time compute	Notes
Pop-slope baseline	0 (ref.)	none	pooled affine calibrator
Naive per-user OLS	+7.02%	none	high-variance at low n_j
Efron–Morris intercept-only EB	+4.70 pp	none	intercept shrinkage only
PEBS (ours)	+8.58% ; +5.88% LOCO	none	stable per-user IDs; Theorem 1
P-GenRM (Zhang et al., 2026)	+8.13% LOCO	prototype clustering + per-prototype OLS	learned prototypes

This appendix expands the claim-bearing diagnostics behind Figure 1: sparse-rater shrinkage, cross-base transfer boundaries, adapter prediction-spread, and the numerical details needed to reproduce the reported CIs.

Trigger coverage criterion. The four-base coherence-only probe was pre-registered with the criterion that any single base inversion bounds the across-family result. The dense panel therefore supports the bounded-transfer claim, not an architecture-universal claim. Two MoE runs (Phi-3.5-MoE-Instruct and Mixtral-8×7B) used a narrower output-projection adapter than the dense-Transformer protocol; both produce negative-direction trained-coherence gains (−59.41% and −60.44%) with narrow prediction spread (0.267 and 0.298). We use these MoE points only as additional boundary evidence consistent with the dense-panel collapse signature, not as full cross-architecture replications.

Calibration diagnostics. The probe measures prediction spread on a 208-row HelpSteer2 slice. The two inversion bases (Llama-3-8B, Yi-1.5-34B) lie below the 0.40 collapse threshold ($\sigma_{\text{pred,coh}}=0.2298, 0.3246$), while Mistral-Small-22B lies above (0.4743); Figure 8 plots the three values. Verbosity-bias and LoRA-capacity alternatives are addressed by the verbosity-only control and the multi-seed Phi-3 re-run. We treat head collapse as an observational signature: the posterior is wide ($P \in [0.30, 0.85]$), and causal mechanism claims require intervention experiments outside this paper.



† : sentinel rule fires (sign-flip on coherence; saturated cells $|v| > 100\%$ extend in colorbar)

Figure 4. Cross-base results on HelpSteer2 (four bases, five attributes). Cell value is the signed gain in per-cent; cell colour encodes one of three outcomes: positive (gain replicates), negative trigger (pre-registered criterion is met), or non-replicating positive. The coherence column shows the two across-family negative outcomes on Llama-3-8B and Yi-1.5-34B that bound the transfer result. The verbosity column is uniformly positive across all four bases, ruling out an attribute-agnostic verbosity bias as the source of the coherence reversal pattern.

Demographic ANOVA on PRISM. An Analysis of Variance of the fitted $(\hat{\alpha}_j, \hat{\beta}_j)$ against six PRISM demographics (age, gender, region, education, political orientation, English fluency) finds only the gender $\rightarrow \hat{\beta}_j$ cell surviving Bonferroni correction, and even there the explained variance is small ($\eta^2 < 0.02$). Demographic grouping cannot replace per-user calibration; the six demographic axes do not jointly recover the per-user shrinkage gain reported in §3.1.

Multi-attribute regression observation on HelpSteer2. We additionally observe the same shrinkage mechanism on a multi-attribute regression problem, where the five HelpSteer2 attribute axes are treated as five pseudo-raters across 1,038 rows. This is an observation about EB-shrinkage stability in a multi-axis regression context, *not* a pluralism claim: the five axes are scoring dimensions, not human annotators with heterogeneous calibrations. The four-seed Qwen-2.5-7B mean is +18.24% relative RMSE reduction [+17.97, +18.51] (across-seed half-width 0.27 pp), reflecting the same calibration-loss-reduction PEBS provides on PRISM applied to a different problem geometry. The result is bound to the Qwen-2.5 family (§3.5, §5); the across-seed half-width is roughly an order of magnitude tighter than the within-seed bootstrap half-width.

HelpSteer2 verbosity per-attribute null. Among the five HelpSteer2 attributes the EB-shrunk arm gains positively on four (helpfulness +6.10%, correctness +7.08%, coherence +41.15%, complexity +30.13%); verbosity straddles zero at -2.74% [-38.04, +27.62] with shrinkage weight $\omega_\beta \approx 0.93$, indicating the attribute is already near-saturated under per-attribute fit and there is little for shrinkage to add. The four-of-five positive pattern rules out an attribute-agnostic verbosity bias as the source of the within-user RMSE gain in §3.1.

Base-model training details. Five-seed Phi-3 re-run gives cross-seed mean +42.18% within 1.05 pp of the single-seed reference, with trained-coherence variance 3.637 pp² versus untrained mean 580.8 pp². Phi-3 verbosity-only control turns trained-verbosity negative to -32.62% while preserving untrained-coherence at +43.18% (Table 2). Qwen2.5-7B-Instruct uses Transformer Reinforcement Learning (TRL) 0.12.2 (von Werra et al., 2020) LoRA $r=32$, $\alpha=16$, lr 10^{-4} , bf16, 1,500 steps, centered-rewards regularizer (Eisenstein et al., 2023), pair accuracy CI [62.74, 65.29], ≈ 75 min H100 80 GB. Bootstrap CIs are 95% BCa (Efron, 1987) with a PRISM 4,000-replicate cluster bootstrap by user (Cameron et al., 2008) and a HelpSteer2 row-cluster. PRISM MoM: $\hat{\tau}_\alpha^2=115.7$, $\hat{\tau}_\beta^2=26.2$, $\hat{\sigma}_\epsilon=23.5$. The 8.58% random-fold-within-user PRISM result attenuates predictably under stricter splits: a strict temporal 80/20 returns +7.55% (30-seed cluster-bootstrap CI

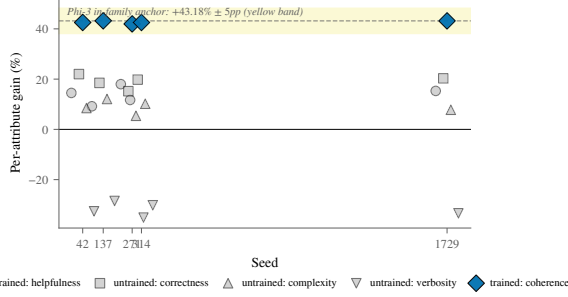


Figure 5. **Phi-3-medium-4k cross-seed scatter.** Each column is one random seed, and each row-coloured line is one HelpSteer2 attribute. Trained-attribute coherence is positive in all five seeds (mean +42.15%, Welch- t 95% CI [+40.10, +44.20]), while untrained attributes scatter more widely.

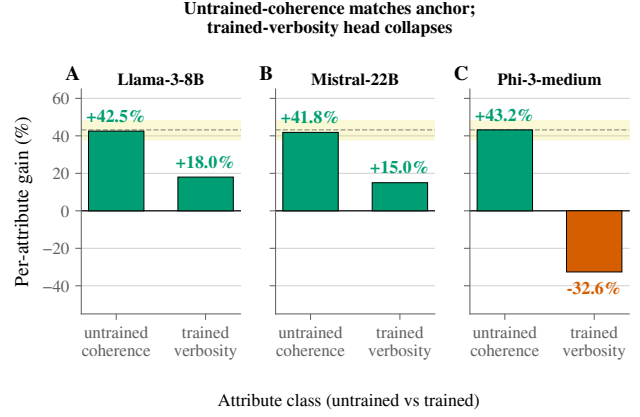


Figure 6. **Verbosity-only control confirms the reversal is attribute-specific, not architecture-wide.** Each cluster shows one base; dark bars are trained-verbosity gain (%), light bars are untrained-coherence gain (%). Across Llama-3-8B, Mistral-Small-22B, and Yi-1.5-34B, untrained-coherence remains positive and within ~ 1 pp of the Phi-3 reference. On Phi-3 the trained-verbosity head turns negative while untrained-coherence still holds, directly ruling out a base-level failure as the cause of the coherence reversal.

[+6.82, +8.71]), cluster-bootstrap-by-user gives +6.96% (BCa [+6.40, +7.56]), and leave-one-conversation-out yields +5.88% (BCa [+5.17, +6.63]); all four exclude zero.

C. PRISM baseline scope

P-GenRM (Zhang et al., 2026) is included as the matched scalar-RMSE baseline and exceeds PEBS in the strict LOCO cell reported in Table 4. Methods whose published protocols optimise a different objective, metric, or feature space are cited in related work but are not reproduced as scalar-RMSE head-to-head rows here, because that would turn a protocol mismatch into an apparent comparison.

D. Dataset cards

This appendix expands the corpora used in §3.4 (the three within-scope continuous-rating corpora) and §3.7 (MultiPref, the theory-predicted scope-limit demonstration corpus), with details on collection, structure, and the operations PEBS requires. None of these corpora is collected by us.

PRISM Alignment corpus (Kirk et al., 2024). A public preference-elicitation corpus with 1,500 unique participants drawn from 75 countries and 24 demographic axes. Each participant has a stable per-annotator ID and contributes multi-turn conversations with multiple model variants, with both turn-level (Likert 0–100) ratings and pairwise preferences. PRISM is the canonical PEBS venue because the per-annotator IDs are stable across conversations, which is required to estimate the per-user (α_j, β_j) random effect. We curate 26,876 preference pairs across 1,391 users with $n_j \geq 3$ pairs from the multi-turn utterance subset, and use an 80/20 stratified-by-user split.

PluriHarms (Li et al., 2026). A harm-rating corpus collecting 15,000 harm ratings on a 0–100 scale from 100 annotators across 150 prompts. Each prompt-response pair is rated by multiple annotators with a stable per-annotator ID. PluriHarms tests whether PEBS’s recipe transfers from preference judgments (PRISM) to a qualitatively different feedback type (single-axis harm rating).

MultiPref (Miranda et al., 2025). A five-point Likert preference corpus in which annotators express preferences with confidence ratings rather than as binary BT-style picks. The per-annotator rating distribution is non-Gaussian, so MultiPref lies outside the Gaussian random-effects regime that PEBS’s MoM estimator assumes. The corpus enters this paper only

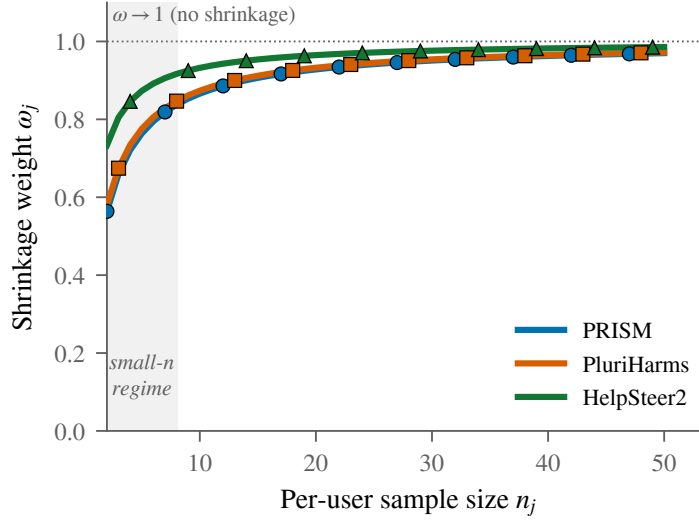


Figure 7. **PEBS automatically down-weights sparse annotators: shrinkage is largest for $n_j \leq 8$ and fades to zero at high n_j with no threshold to tune.** The closed-form weight $\omega_j = \tau^2 / (\tau^2 + V_j)$ governs how much PEBS trusts each rater’s own calibrator vs. the population mean. Three illustrative populations (PRISM, PluriHarms, HelpSteer2) are plotted against per-user sample sizes n_j and within-rater noise $V_j \propto 1/n_j$; the shaded zone ($n_j \leq 8$) is where shrinkage carries the most weight. ω_j asymptotes to 1 (no shrinkage) as n_j grows, so dense annotators are trusted fully without any manual cutoff.

as the theory-predicted scope-limit demonstration discussed in §3.7; the negative-control framing, the predicted-versus-observed numerical gap, and the principled Beta-Binomial or Student- t_ν random-effects extension are all documented there.

HelpSteer2 attribute-as-rater recast (Wang et al., 2024). HelpSteer2 provides five scalar attribute ratings per prompt-response pair (helpfulness, correctness, coherence, complexity, verbosity) on a 0–4 scale, from a panel of human annotators whose individual identities are not released. Because PEBS requires per-rater data, we re-cast the corpus by treating the five attribute axes themselves as five *pseudo-raters*: each row contributes one rating from each axis, so a single prompt-response pair is rated by all five attribute “raters”. The HelpSteer2 attribute-as-rater protocol uses 1,038 rows. The cross-family probes of §3.5 train a coherence-only LoRA adapter (loss masked to the coherence axis) and a verbosity-only counterfactual (loss masked to verbosity) on each of five base architectures.

Forecast companion corpora. OASST2-author (Köpf et al., 2023) and SHP-subreddit (Ethayarajh et al., 2022) are open preference corpora used as additional Morris g -function validation points. We use stable author- or subreddit-level grouping variables. We do not run the full PEBS pipeline on these corpora; only the g -function pilot estimate.

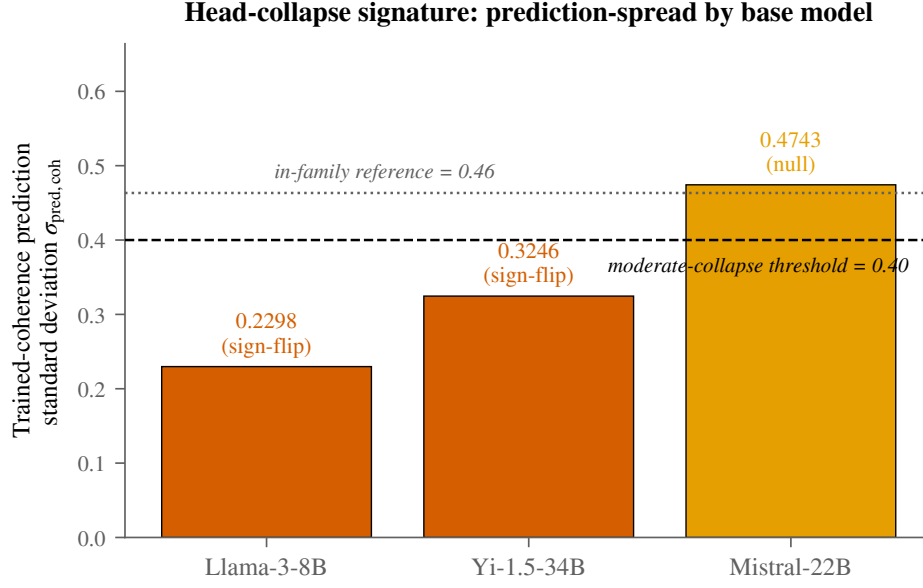


Figure 8. Adapter prediction-spread ($\sigma_{\text{pred, coh}}$) for three across-family bases, against the 0.40 collapse threshold. The two inversion bases (Llama-3-8B, Yi-1.5-34B) fall below the threshold; the null base (Mistral-Small-22B) lies above. Lower $\sigma_{\text{pred, coh}}$ indicates tighter clustering of adapter outputs around one or two rating values, consistent with a head-collapsed adapter setting. We treat the signature as observational rather than causal.

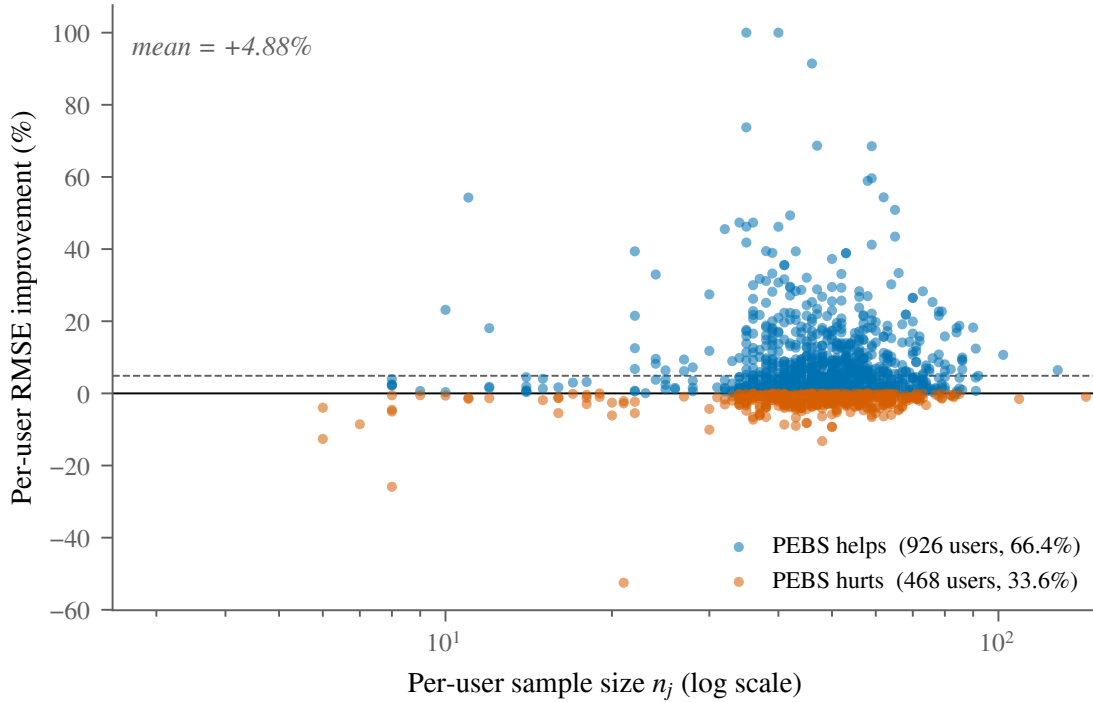


Figure 9. Per-user RMSE improvement scatter on PRISM ($N=1,391$ users), illustrating the minority-rater trade-off of EB shrinkage. Each point is one user; the x -axis is per-user sample size n_j (log scale) and the y -axis is the per-user RMSE improvement (pop-slope minus PEBS-shrunk, as a percentage of pop-slope RMSE). Blue points (66.4% of users) are helped by PEBS; vermilion points (33.6%) are hurt. Low- n_j users are shrunk most aggressively ($\omega_j \rightarrow 0$ as $n_j \rightarrow 0$) and show the widest spread in improvement, consistent with the standard EB trade-off: optimal under the prior but wrong for the rare true-extreme rater.